

Duplicated Virtual Co-embodiment Displaying Out-of-view Movements for Tennis Training

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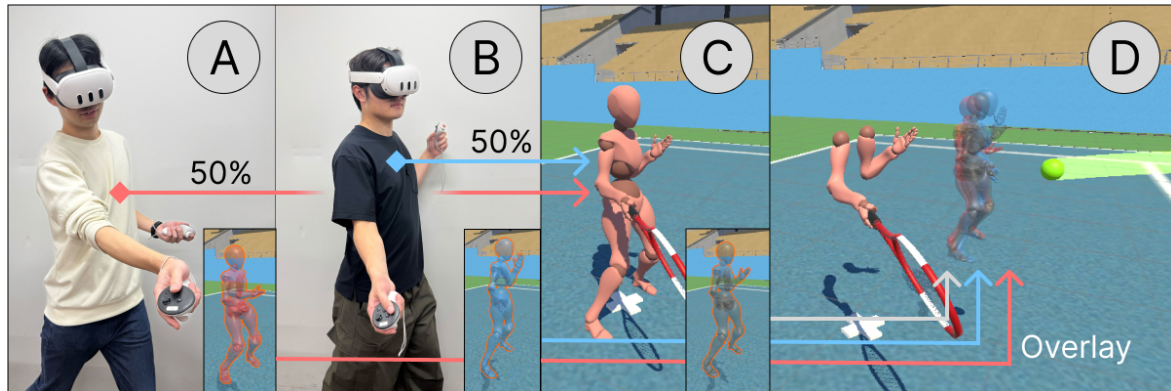


Figure 1: Concept of tennis training with duplicated virtual co-embodiment. (A) Teacher and the duplicated avatar (red). (B) Learner and the duplicated avatar (blue). (C) Co-embodied avatar and the duplicated avatar (black). (D) The training system in action, where the original co-embodied avatar's arms are visible. Three duplicated avatars are overlaid and displayed in front of the user.

ABSTRACT

Motor skill instruction in virtual environments allows for the presentation of different body information from physical environments, thereby supporting more efficient learning. Virtual Co-embodiment, in which a learner and teacher share control of a single avatar, has demonstrated promising results in motor skill acquisition. However, existing methods are typically limited to movements within the user's field of view, making their applicability to sports involving out-of-view movements uncertain. To address this issue, we developed a system that displays duplicated avatars in front of the user, allowing the user to observe and adjust their movements more easily. In this study, we focused on tennis return shots and displayed the teacher's, the learner's, and the co-embodied avatars, enabling users to recognize and correct differences in their movements through virtual co-embodiment. We conducted a user study on hitting back a tennis ball. We found that providing learners with too much information at once makes correcting their movements difficult. The comments also suggested that reducing the amount of information shown to learners or slowing down their movements may be necessary.

Index Terms: Human-centered Computing, Visualization, Motor Learning, Education, Virtual Co-embodiment

1 INTRODUCTION

Observing expert's movements and integrating them into their own is a key factor in improving sports performance. This aspect is par-

ticularly important for beginners, who may have trouble adapting expert movements to their bodies, delaying progress by exploring their movements. To address this issue, methods that incorporate multiple sensory modalities, such as visual, haptic, and somatosensory feedback, are expected to facilitate more efficient motor skill learning. For example, Nishida et al.[8] used direct physical interventions to help learners internalize movement, while Nijima et al.[7] employed electromyographic stimulation to boost motor skills. However, these interventions require specialized equipment, limiting their accessibility to broader users. In contrast, recent advances in motor skill learning through virtual environments utilize easily obtainable head-mounted displays (HMDs), making them more accessible for a wide range of users.

Virtual Reality (VR) technologies enable information presentation methods that would be difficult or impossible to achieve in physical environments. Yang et al.[10] developed "Just Follow Me," which overlays the teacher's avatar onto the learner's in virtual environments, facilitating movement imitation from a first-person perspective. While this method provides clear visual guidance, learners may develop overreliance on tracing teacher movements, potentially hampering skill retention during independent practice. To address this limitation, researchers introduced "virtual co-embodiment," where teacher and learner jointly control a single virtual avatar ("co-embodied avatar"). This approach allows learners to experience the teacher's movements while actively practicing in real time[1, 2]. Research by Kodama et al.[6] demonstrated that co-embodied avatar training yielded superior skill improvement compared to both independent practice and the "Just Follow Me" method, indicating virtual co-embodiment's potential for effective VR-based motor skill transfer.

Prior research has focused on movements within the HMD's field of view. However, in physical sports environments, athletes rarely maintain complete visual awareness of their entire body, instead relying on somatosensory and haptic feedback from body parts outside their direct vision. Researchers have explored methods of projecting body movements into the user's field of view

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to address this limitation. Ikeda et al.[5] developed a system that projects both teacher and learner shadows during golf swings, enabling learners to visualize their form and plan subsequent movements more effectively. Similarly, Han et al.[3] enhanced movement correction by positioning virtual avatars around the learner's body. While methods for projecting learners' bodies into their field of view exist, research remains limited on interactive techniques—such as virtual co-embodiment—enabling teachers' real-time, full-body movement correction. To address this gap, we developed a VR tennis system displaying duplicated versions of individual avatars (teacher and learner) and the co-embodied avatar in front of users. This configuration facilitates real-time teacher intervention while enabling users to identify and correct discrepancies in out-of-view movements. We present the system's design architecture and evaluate its effectiveness through user study findings.

2 SYSTEM DESIGN

In our system, a duplicated avatar is displayed in front of the learner so that they can improve their movements while still receiving support from the teacher. This setup extends the benefits of virtual co-embodiment even to movements that lie outside the learner's normal field of view. The virtual co-embodiment in this system is split evenly between the teacher and the learner, with each controlling 50% of the movement. While this shared control may lead to conflicts in their intended movements, it ultimately facilitates learning by helping the learner understand how his movement differs from the teacher's, and enables the teacher to identify the learner's problematic movement through direct physical experience.

2.1 Virtual Co-Embodiment

This paper implements virtual co-embodiment using a weighted-average-based method[1]. First, we obtain the position and orientation of each user's head and hands from the HMDs and controllers worn by both users. We then calculate the co-embodied avatar's head and hand positions and orientations by taking a weighted average of these measurements. The weighted averaging process employs linear interpolation for positions (represented as 3-dimensional vectors) and spherical linear interpolation for orientations (represented as quaternions). Finally, we compute the remaining posture of both users' avatars and the co-embodied avatar through inverse kinematics¹ (see Figure 1(A), (B), and (C)).

We implement the multi-user functionality through a server-client architecture using Unity Netcode for GameObjects². We use a gaming laptop as the server and two Meta Quest 3³ headsets as clients. We rely on wireless local network communication, allowing users to experience virtual co-embodiment without physical tethering constraints. We can dynamically adjust the weights from the server-side interface.

We place three avatars in front of the user: a duplicated co-embodied avatar and two avatars corresponding to each user. By observing these three avatars, the user can see how their movements differ depending on the user. We display only the arms of the original co-embodied avatar to ensure visibility in the user study. We present the virtual environment in Figure 1(D).

2.2 Tennis Game in Virtual Reality

We implemented the tennis game with the Open Video Game Library (see Figure 1(D))[4]. In this game, users use a racket attached to the right hand of the co-embodied avatar to hit back the ball served from the opponent's court. We marked the ball with a green trajectory so users can easily track it. Players can serve the ball from the server-side interface (see Figure 2).

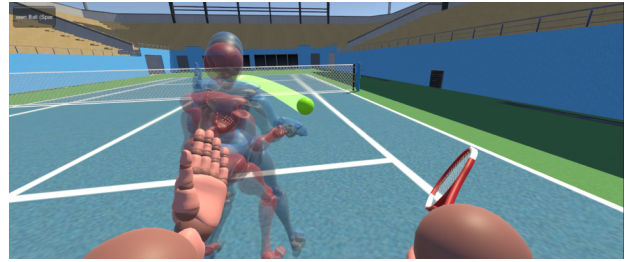


Figure 2: The virtual environment provides a first-person perspective where three transparent avatars—teacher (red), learner (blue), and co-embodied (black)—are overlaid and displayed in front of the user. Users can simultaneously observe these avatars along with the virtual environment elements, including the tennis balls and racket.

3 USER STUDY

This study has two objectives: to simply validate the effectiveness of our system for learning movements outside the visual field, and to explore potential learning methodology improvements based on user feedback. Given the exploratory and preliminary nature of this investigation, we did not seek formal ethical approval, but we did obtain feedback from VR developers within our professional network who have extensive experience in the field. In this study, two participants were recruited: a 23-year-old male tennis teacher with 10 years of experience and a 24-year-old male learner with almost no tennis experience. First, they moved in the physical environment to be on the cross markers placed in the virtual environment. Next, they independently practiced returning a ball. Then, they performed the same returning task with a co-embodied avatar. After the tasks, they were interviewed about their impressions of the experience.

The teacher's feedback expressed that physically holding the learner's arm and correcting the swing (commonly done in face-to-face lessons) can be sufficiently replicated in VR. The teacher also mentioned the potential of displaying three duplicated avatars to improve learning efficiency. The learner can compare their swing to the correct form. The teacher emphasized the importance of movement initiation and footwork timing since these factors significantly affect tennis returns. However, participants expressed concerns that real-time corrections and following movements based on virtual co-embodiment may not be fully perceptible for high-speed swings.

From the learner's perspective, the primary difficulty was maintaining focus on the avatar while concentrating on the ball. In addition, the simultaneous display of multiple avatars created issues with visual clarity, raising the possibility that simply comparing the teacher's and learner's avatars might ultimately be sufficient for effective instruction. Furthermore, if the teacher and learner differ significantly in physical build or swing speed, the co-embodiment avatar may fail to convey genuine physical sensations accurately.

Participants suggested potential improvements, noting that system constraints limit synchronization accuracy for high-speed, practical play. However, co-embodiment-based instruction seems to be highly effective at slower speeds, where the focus is on carefully checking one's form. From the teacher's viewpoint, an approach that divides coaching into two stages (a phase focusing on thorough swing corrections followed by a phase approaching normal swing speeds) is considered promising for teaching scenarios. Meanwhile, from the learner's perspective, it is not strictly necessary to use co-embodiment at all times; in many situations, comparing the teacher's and learner's avatars while communicating verbally may be sufficient for achieving satisfactory learning outcomes.

¹ <https://assetstore.unity.com/packages/tools/animation/final-ik-14290>

² <https://unity.com/en/products/netcode>

³ <https://www.meta.com/en/quest/quest-3>

4 DISCUSSION

We discuss findings from our user study and focus on two factors that affect motion learning in racket sports: visibility and ball speed. First, visibility can become problematic when the teacher and the learner view a co-embodied avatar simultaneously. Overlapping avatars can increase the learner's cognitive load and draw attention away from the ball. One solution is to separate the display and feedback for the teacher and the learner. This approach helps reduce mental overload and supports accurate motion learning. Second, ball speed becomes a challenge when multiple people train together. Learners may have little time to observe and correct their form in fast-paced conditions. A slowdown mechanism at key moments could help. Noguchi et al. [9] demonstrated that using a "hit stop" between two people does not create a sense of discomfort. We propose adapting their idea to co-embodiment training. This method would allow learners to scrutinize the critical moment of impact and refine their form more effectively. Our user study reveals that movement initiation and footwork timing strongly affect tennis returns. However, the current virtual co-embodiment system fails to reflect these factors. We can integrate them to help learners grasp full-body movements more comprehensively. Kodama et al.[6] reported that co-embodiment could outperform a "Just Follow Me"[10] approach. However, our feedback indicates that co-embodiment may not suit sports requiring rapid or dynamic actions. At least in a weighted-average model, virtual co-embodiment overlooks time elements and struggles to adapt to high-speed or momentary movements. We must tailor virtual co-embodiment to each sport's needs.

5 CONCLUSION

In this study, we placed the co-embodied avatar within the user's field of view, enabling the teacher and learner to perform tennis return motions collaboratively. While the presentation method showed potential for providing a sense of embodied collaboration beyond what is possible in traditional face-to-face coaching or video-based comparison, actual user experiences indicated that issues related to visibility and difficulty in keeping up with high-speed movements made it hard to appreciate the advantages of co-embodiment fully. Our findings suggest that reducing cognitive load and improving synchronization accuracy in fast-paced actions—through techniques such as refining multi-avatar displays or introducing momentary slowdown at the point of impact—could lead to more effective learning. Moreover, designing a system that allows users to switch between co-embodiment control and ghost-style visualization may facilitate step-by-step support, ranging from initial skill acquisition to near-realistic speeds, enabling flexible learning of various form corrections and timing adjustments. Addressing these design challenges can further expand the application of co-embodiment in broader sports movement training contexts.

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